SpendGini

Offline Personal Expense Intelligence

PROJECT DOCUMENTATION & SOLUTION OVERVIEW

Your Spending. Your Genie. Your Control.

Executive Summary

SpendGini is a privacy-first personal expense management application designed to help individuals record expenses, understand spending patterns, manage recurring obligations, and ask questions about their recorded spending. The visible application is delivered through a local browser interface and explicitly presents itself as offline and stored on the user's computer.

Its central design principle is practical: provide useful personal finance visibility without requiring users to send their transaction history to a cloud service. The project combines structured expense entry, dashboard reporting, recurring schedules, a local receipt-reading pilot, and an offline query interface called Ask Gini.

Document purpose

This document describes the product vision, demonstrated capabilities, conceptual workflows, design choices, risks, and a realistic roadmap. It is suitable for portfolio review and as a foundation for a user manual and executive presentation.

Evidence and boundaries

The supplied evidence comprises five application screenshot collections (13 pages total): Dashboard, Expenses, Recurring, Receipt Scan, and Ask Gini. Statements about visible interface behavior are grounded in those screens.

1. Business Context & Problem

Personal expenses are often fragmented across receipts, manual notes, recurring commitments, and bank transactions. Users can lose sight of what they spent, where they spent it, and how month-to-month patterns change. Tools that depend on cloud synchronization can introduce unwanted data-sharing concerns.

Problems addressed

- Difficulty consolidating everyday purchases into a consistent expense record.
- Limited visibility into monthly totals, daily trends, merchants, and categories.

- Confusion between scheduled bills and payments actually made.
- Time-consuming entry of paper receipts, especially item-level detail.
- Need for quick answers without uploading private financial history.

Product proposition

SpendGini organizes spending into a single locally operated experience, with an emphasis on user control, clarity, and incremental automation. It is an expense-tracking tool—not a banking platform, income-management system, or verified financial-advice service.

2. Objectives, Scope & Success Criteria

Objective	Expected user outcome
Record spending	Capture a dated, categorized transaction with merchant and amount.
Understand spending	Review monthly totals, transaction counts and comparisons.
Track commitments	Maintain recurring schedules separately from recorded payments.
Reduce receipt entry	Use local OCR as an editable starting point for review.
Ask questions	Retrieve spending answers from saved expenses without online AI.
Preserve privacy	Keep expense processing and storage on the local computer as described in the UI.

In scope

Expense records, dashboard analytics, recurring expense schedules, receipt OCR pilot, and Ask Gini queries.

Out of scope / deferred

Income and savings management are outside the intended product scope. Budget planning and tracking, and robust reconstruction of long or multi-photo receipts, remain deferred improvements. No claim is made that these are completed features.

Success measures to validate

- Correctness of monthly spending totals and transaction counts.
- No double counting between recurring schedules and posted payments.
- Accurate filtering by date, merchant and category.
- User-reviewed OCR results before they become saved expenses.
- Consistent answers from Ask Gini and their supporting transactions.
- Successful local operation without an internet dependency for core workflows.

3. Application Experience & Navigation

The visible left navigation presents Dashboard, Expenses, Recurring, Receipts, Ask Gini and Settings. The interface uses a consistent sidebar and a restrained card-based layout to help users move between capture, monitoring and exploration.

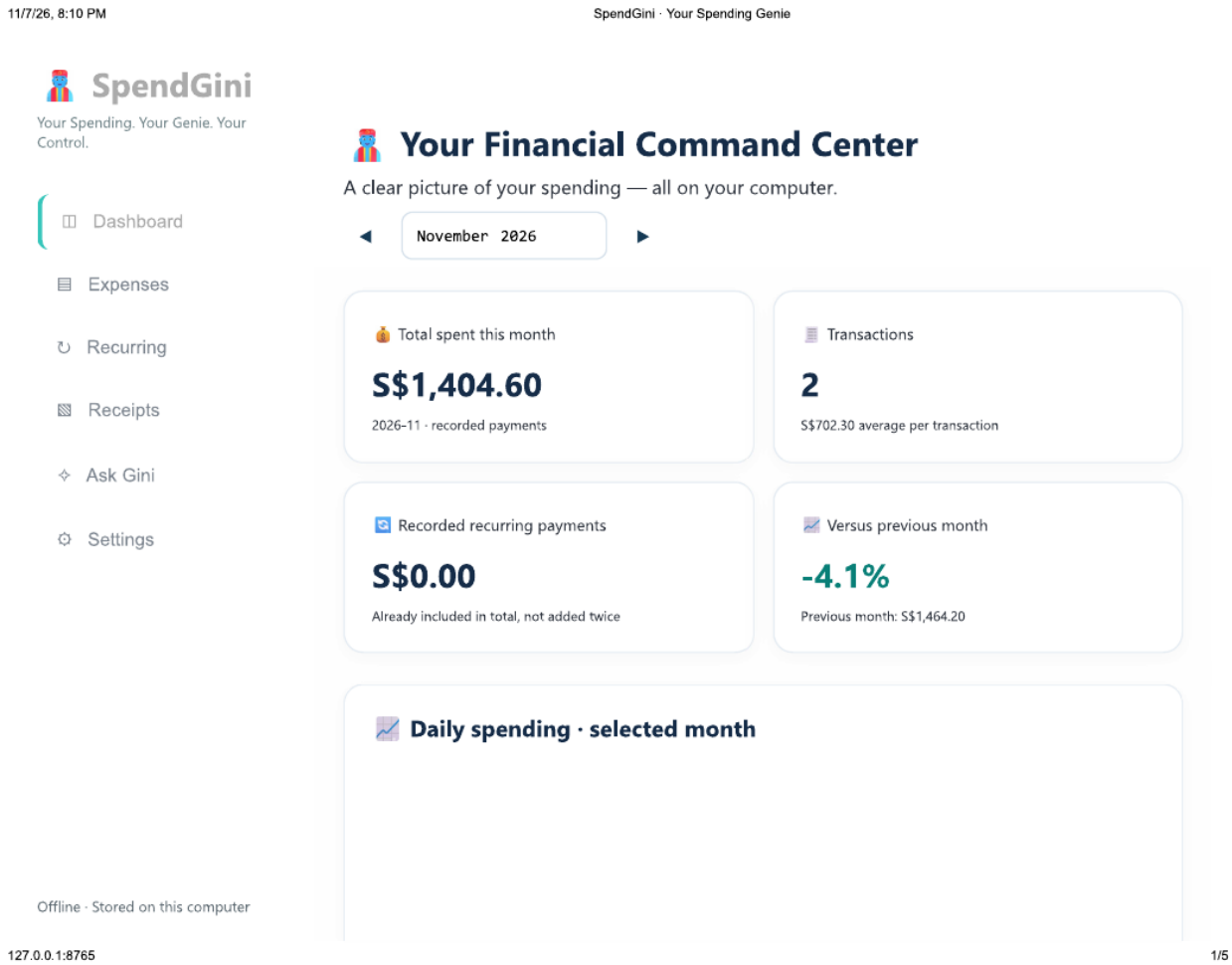


Figure 1. Financial Command Center: monthly summary and spending trend area (supplied screenshot).

Design

The application makes the current month explicit, displays currency in SGD, and separates headline metrics into individual cards. The sidebar includes the message “Offline · Stored on this computer.” These are visible interface statements, not independent security-audit findings.

4. Financial Command Center

The Dashboard is the primary monitoring view. In the provided example, the selected month is November 2026. The cards show total spending, number of transactions, recorded recurring payments, and percentage change against the prior month.

Displayed metric	Interpretation
Total spent this month	Sum of recorded expense payments for the selected month.
Transactions	Count of recorded transactions, with average amount displayed.
Recorded recurring payments	Recurring payments already recorded; included in the total, not added again.
Versus previous month	Relative comparison with the prior month.
Daily spending	Time-based visualization for the selected month.

Example visible in screenshot

November 2026 shows S$1,404.60 across 2 transactions, with an average of S$702.30. The displayed comparison is −4.1% against the previous month. These figures are screenshot examples and must not be interpreted as representative user behavior or validated production statistics.

Analytical value

The dashboard moves beyond simple recordkeeping by placing transactions in context: month-to-month change, payment composition, and daily patterns. This supports self-service spending awareness.

5. Expense Management

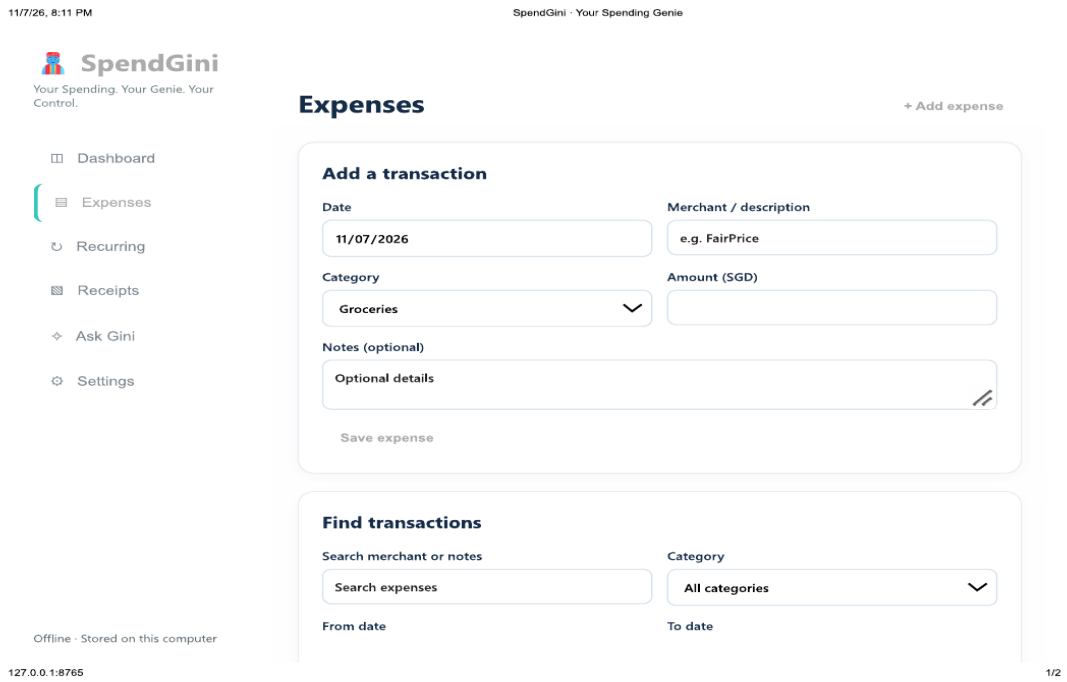


Figure 2. Expenses screen showing transaction entry and search controls.

The expense form visibly supports date, merchant/description, category, amount in SGD, and optional notes. A separate Find transactions area provides text search and category and date filtering.

Core workflow

- Open Expenses and select or expand Add a transaction.
- Enter the date, merchant/description, category and amount.
- Optionally add explanatory notes.
- Save the transaction; verify it appears in the transaction list and relevant dashboard period.
- Use search, category and date filters to find previously recorded entries.

Data quality considerations

The application should validate dates and positive amounts, avoid accidental duplicate entries, and make corrections traceable. These are recommended controls; the supplied screens do not establish that each validation rule is implemented.

6. Recurring Expenses

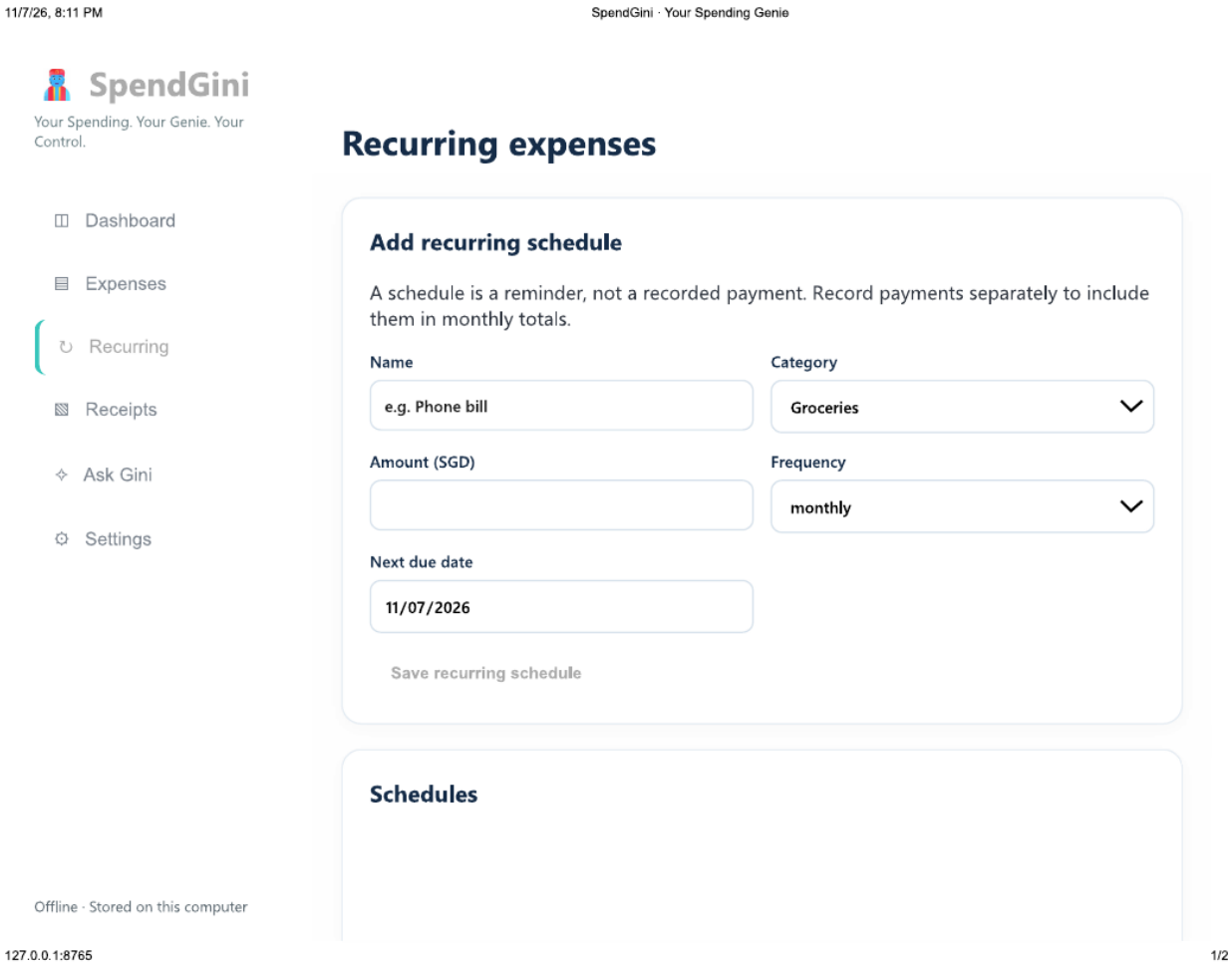


Figure 3. Recurring schedule entry, with explicit separation of schedules and actual payments.

The recurring view supports a named schedule, category, amount, frequency, and next due date. The interface explicitly explains: “A schedule is a reminder, not a recorded payment. Record payments separately to include them in monthly totals.”

Why this matters

Keeping planned obligations distinct from recorded spending is essential for accurate expense analytics. A phone bill scheduled for next week should not inflate the current month’s actual expenditure until the user records the payment.

Recommended user workflow

- Create a recurring schedule for a subscription or bill.
- Review due dates and upcoming commitments.
- When payment occurs, record the actual transaction.
- Verify the transaction is counted once in monthly spending.

7. Receipt Magic — Local OCR Pilot



Figure 4. Receipt Magic v0.3A: local OCR and editable extraction interface.

Receipt Magic screen supports selecting one or more overlapping receipt photos in top-to-bottom order, with JPG, PNG or WEBP input, a stated 8 MB per image limit, and a maximum of eight photos. It offers Quick Scan and Enhanced Scan, as well as positional OCR diagnostics.

Visible privacy and review controls

- The interface states OCR runs locally using Tesseract.
- The interface states images and raw OCR text are not saved.
- Only explicitly approved expense and item records are saved locally.
- Extracted OCR text is editable before approval.

Pilot limitations

Long receipts, overlapping images, text ordering, and item-level reconstruction can be error-prone. This workflow should be described as assisted capture with human review—not guaranteed automatic extraction. Improvements to complex multi-photo receipt reconstruction are deferred.

8. Ask Gini — Offline Spending Questions

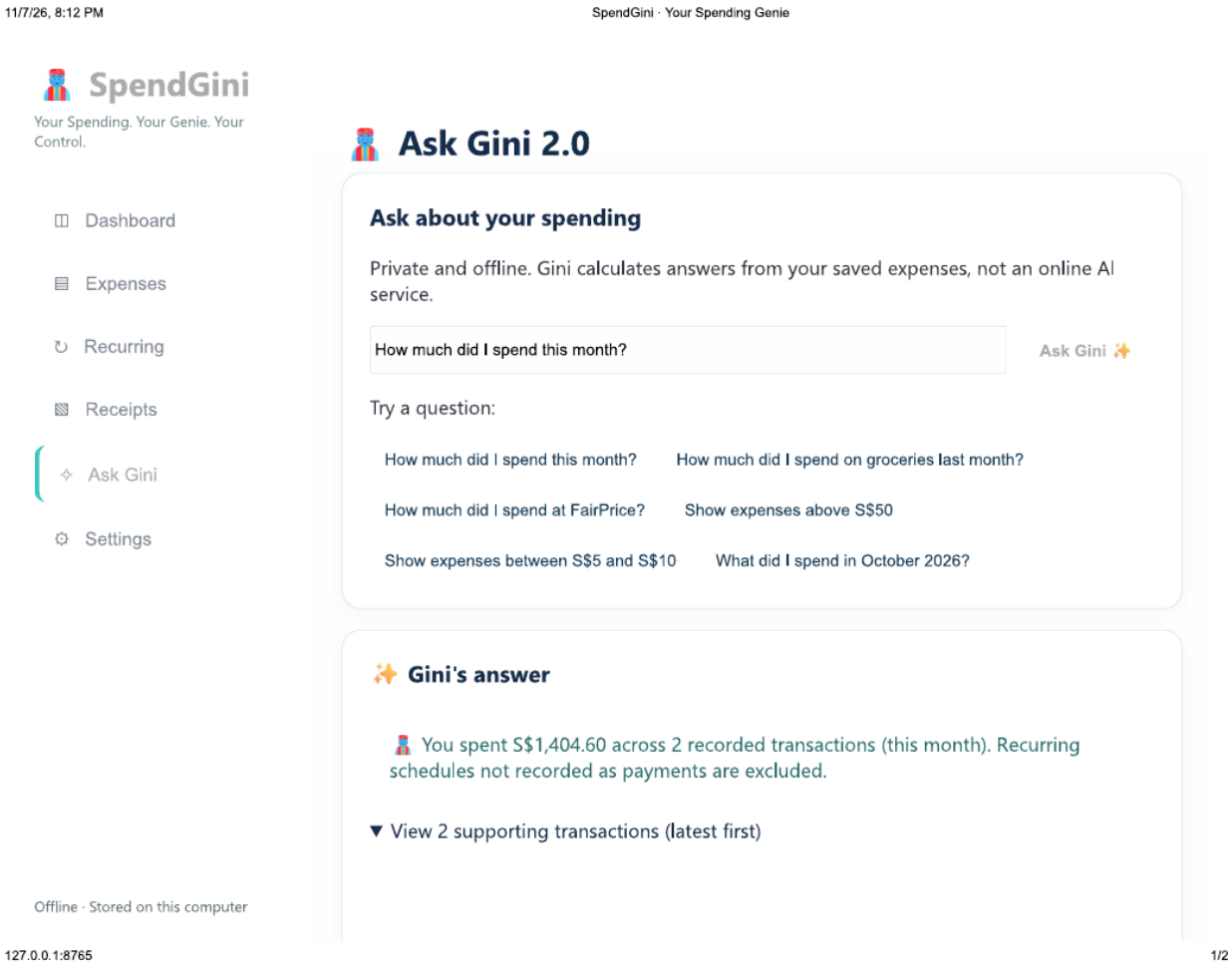


Figure 5. Ask Gini 2.0: question input, calculated answer, and supporting transactions.

Ask Gini provides a conversational-style interface for querying recorded expenses. The screenshot states: “Gini calculates answers from your saved expenses, not an online AI service.” Suggested prompts include spending this month, groceries last month, spending at a named merchant, amounts above a threshold, and expenses within a range.

Illustrative output

In the supplied screen, Ask Gini returns S$1,404.60 across two recorded transactions for the current month and provides an expandable supporting-transaction list. It explicitly excludes recurring schedules that have not been recorded as payments.

Trust and transparency

The ability to inspect supporting transactions is a valuable design feature: users can trace an answer to the underlying records. Ask Gini should be described as an offline, rule/data-driven spending query experience; the screenshots do not establish that a large language model is used.

9. Conceptual Solution Architecture

The following is a functional architecture inferred from the user-visible workflows. It is not a verified source-code or deployment diagram.

Layer	Responsibility
Local user interface	Dashboard, Expenses, Recurring, Receipts, Ask Gini, Settings.
Expense workflows	Manual entry, validation, search and categorization.
Recurring workflows	Maintain schedules; distinguish reminders from recorded payments.
Receipt processing	Local OCR, editable extraction, user approval.
Analytics and query	Monthly metrics, trend views and offline question answering.
Local persistence	Save approved expense/item data on the computer (per UI).

Conceptual data entities

- Expense: date, merchant, category, amount, optional notes.
- Recurring schedule: name, category, expected amount, frequency, next due date.
- Receipt-derived item: description and item-level fields, subject to user approval.
- Category and merchant: classification and filtering dimensions.

Privacy posture

Local operation minimizes the need to transmit personal financial information to external services.

10. Quality Assurance & Validation Approach

A reliable personal finance application must prioritize correctness over visual polish. The following checklist is a proposed test plan, not a claim that all tests have passed.

Test area	Representative verification
Manual entry	Save valid expense; reject missing/invalid values.
Dashboard totals	Reconcile displayed totals with underlying recorded transactions.
Date filtering	Check month boundaries and cross-year navigation.
Recurring	Verify schedule alone does not affect actual spending.
Receipt OCR	Compare extracted text/items with source image before approval.
Ask Gini	Cross-check answer against supporting transaction list.
Offline use	Repeat core flows with network access disabled.
Data safety	Test backup/restore, corruption handling and deletion behavior if supported.

Current validation status

Screenshots demonstrate interface availability and example outputs, but they are not a substitute for a repeatable automated test suite or formal security review. No performance, accuracy, or security certification is asserted.

11. Risks, Constraints & Mitigations

Risk	Potential impact	Mitigation direction
OCR misreads	Incorrect items or amounts	Editable text, review and approval before save.
Duplicate transactions	Overstated spending	Duplicate detection and user confirmation.
Local device loss	Loss of expense history	Documented backup and restore capability.
Ambiguous queries	Misleading answers	Show filters and supporting transactions.
Data security	Exposure on shared devices	Evaluate encryption, access

		controls and secure storage.
Scope expansion	Delayed delivery	Stage features and maintain a backlog.

12. Development Roadmap

Stage	Focus	Status / evidence
Current experience	Dashboard, manual expenses, recurring schedules, Ask Gini	Visible in supplied screens.
Receipt capture	Quick/Enhanced Scan, positional OCR diagnostics	Pilot visible; complex receipts need improvement.
OCR hardening	Long and multi-photo receipt reconstruction	Deferred backlog.
Budget planning	Monthly/category budgets and comparisons	Deferred backlog.
Reliability	Data validation, repeatable tests, backup/restore	Recommended; implementation unverified.
Distribution	Installer/package and onboarding documentation	Future documentation and validation.

13. Conclusion

SpendGini is an offline-first expense intelligence application that connects everyday expense capture with meaningful financial visibility. Its most distinctive design choices are the separation of scheduled and recorded spending, locally processed receipt OCR with explicit user approval, and Ask Gini answers linked to supporting transactions.